

Digital Maturity Model Survey Form

A printable assessment form for reviewing digital capability across rural drinking water services.

Current model v2.3 · 36 capabilities · 6 layers · 0–4 scale

Assessment details

State / UT

Assessment date

Assessor name

Designation

Department / agency

Review round

How to complete this form

1. Read each capability question and select one maturity score from 0 to 4.
2. Base the score on what is in routine use at the State/UT scale, not only on a sanctioned plan or pilot.
3. Use the evidence / notes area to record the system name, coverage, source, or actions needed.
4. Answer every capability. The complete assessment contains 36 responses and has a maximum score of 144.

0

Does not exist
No digital system for this capability. Handled on paper, in spreadsheets, or not at all.

1

Under development
A system has been sanctioned, procured or is in development. Nothing is in productive use yet.

2

Pilot complete
Live and working with real data in a limited setting — one or a few districts, or a small user group — for at least one full reporting cycle. Not yet rolled out further.

3

Functional at limited scale
In routine production use across part of the state by the intended users. Gaps remain — some districts still manual, or some workflow steps still offline.

4

Fully functional at state scale
In routine use across effectively all districts by effectively all intended users. The data is authoritative, no parallel manual register is maintained, and the system is owned and monitored.

Citizens

Digital services reaching consumers and households directly

1 Grievance & Feedback

Whether the State has the digital capability to systematically collect, analyze, and act upon citizen grievance & feedback regarding rural drinking water services.

Includes: Feedback capture · Sentiment analysis · Closure of loop

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

2 Consumer Registry

Whether the State has a digital registry of FHTC / consumers with a unique identifier, and whether this database is authoritative, complete, up-to-date, interoperable and actively used.

Includes: Unique consumer ID · Household mapping · Interoperability · Active use

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

3 Connection Lifecycle Management

Whether the State effectively manages the entire lifecycle of a Functional Household Tap Connection (FHTC): from application through installation, modification, transfer, suspension and disconnection.

Includes: Application · Installation · Transfer · Suspension · Disconnection

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Citizens

Digital services reaching consumers and households directly

4 Billing & Payments

Whether the state has the capability to digitally view bills, receive payment reminders, make online payments through multiple channels, access payment history and receive digital receipts.

Includes: Bill view · Reminders · Online payment · Payment history · Digital receipts

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

5 Notifications & Alerts

Whether the system has the capability to notify citizens before they need to ask — planned supply interruptions, water quality advisories, bill due reminders, complaint status updates or restoration notices.

Includes: Supply interruptions · Quality advisories · Bill reminders · Complaint status

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

6 Citizen Engagement & Participation

Whether the State provides digital mechanisms that enable citizens and communities to actively participate in governance, monitoring, planning and decision-making.

Includes: Participation · Community monitoring · Planning inputs

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Layer 1 subtotal — Citizens

Sum of 6 ratings

Score: _____ / 24

Frontline Workers

Digital tools used by field staff for monitoring and reporting

7 Water Quantity & Regularity

Whether the system has the capability to enable frontline workers to accurately measure, digitally report and validate daily water supply data in a timely manner using standardized processes and digital tools.

Includes: Daily reporting · Validation · Integration · Governance

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

8 FTK Testing

Whether the system has the capability to enable frontline workers to conduct Field Test Kit (FTK) tests, digitally report water quality observations, identify potential quality issues and support routine village-level surveillance.

Includes: FTK testing · Digital reporting · Issue identification

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

9 Repair and Maintenance

Whether the system has the capability to effectively monitor minor maintenance and repair activities to ensure timely completion, quality of repairs, asset uptime, and continuous improvement.

Includes: Maintenance logs · O&M expenditure · Collections

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Frontline Workers

Digital tools used by field staff for monitoring and reporting

10 Community Engagement

Whether the system has the capability to enable frontline workers to digitally plan, conduct, record and monitor community mobilization and IEC activities that promote safe water practices and sustainable operation.

Includes: IEC planning · Activity records · Monitoring

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

11 Financial Management

Whether the system has the capability to enable frontline workers to digitally record O&M expenditure, collections and village-level financial management.

Includes: Field collection · Inspections · Surveys · Compliance records

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

12 Digital Work Assistance

Whether the State has the capability to digitally empower frontline workers through mobile applications, role-based dashboards, contextual digital assistants, training, knowledge resources and alerts.

Includes: Mobile apps · Role dashboards · Training · Alerts

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Layer 2 subtotal — Frontline Workers

Sum of 6 ratings

Score: _____ / 24

Department

Digital systems for planning, finance, and governance at department level

13 Infrastructure Planning & Asset Governance

Whether the State has the capability to digitally plan, design, govern and manage rural water infrastructure throughout its lifecycle, including DPR preparation, asset inventory, commissioning and lifecycle management.

Includes: Asset Management · Inventory · BOQ · DPR · Asset Registry

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

14 Vendor & Financial Management

Whether the State has the capability to digitally manage procurement, contracts, work orders, invoices, payments and financial transactions, ensuring financial accountability and integration with government financial systems.

Includes: Vendor Management · Contracts · Work Orders · Invoices · Payments · IFMS

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

15 Water Resource & Sustainability Management

Whether the State has the capability to digitally assess, monitor and manage water sources, hydrogeology and source sustainability to ensure long-term availability and resilience.

Includes: Source Sustainability · Source Geology · Water Audits

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Department

Digital systems for planning, finance, and governance at department level

16 Water Service Intelligence

Whether the State has the capability to digitally monitor, analyse and optimise rural water service delivery using operational data, dashboards, analytics, predictive insights and decision-support tools.

Includes: BI · MIS · Dashboards · Predictive Maintenance · Scheme Risk Forecasting

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

17 Water Quality Governance

Whether the State has the capability to digitally monitor, analyse and govern drinking water quality through laboratory management, surveillance, compliance monitoring and timely corrective actions.

Includes: LIMS · Water Quality surveillance

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

18 Engineering & Network Intelligence

Whether the State has the capability to digitally model, monitor and optimise water supply infrastructure using engineering tools, spatial intelligence, network modelling and operational telemetry.

Includes: GIS · Geo-tagging · Hydraulic Modeling · IoT · SCADA

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Department

Digital systems for planning, finance, and governance at department level

19

Project Delivery & Service Quality

Whether the State has the capability to digitally plan, execute, monitor and validate rural water infrastructure projects, ensuring timely completion, adherence to quality standards, and compliance with contractual and technical requirements.

Includes: Planning · Execution · Quality checks · Compliance

☐

0

Does not exist

☐

1

Under development

☐

2

Pilot complete

☐

3

Functional at limited scale

☐

4

Fully functional at state scale

Evidence / notes

20

Operations & Resource Management

Whether the State has the capability to digitally plan, allocate and monitor operational resources, including workforce, equipment, materials, consumables and finances, to support efficient O&M of water supply schemes.

Includes: Workforce · Equipment · Materials · Finances

☐

0

Does not exist

☐

1

Under development

☐

2

Pilot complete

☐

3

Functional at limited scale

☐

4

Fully functional at state scale

Evidence / notes

Layer 3 subtotal — Department

Sum of 8 ratings

Score: _____ / 32

State Functionaries

Cross-department coordination and strategic decision-making

21 Interdepartmental Coordination & Collaboration

Whether the State has the capability to digitally coordinate, collaborate and exchange information across departments to enable seamless execution of programmes and coordinated governance of rural drinking water services.

Includes: Data exchange · Workflow · Coordination

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

22 Integrated Planning & Programme Governance

Whether the State has the capability to digitally support joint planning, approvals, programme monitoring and governance across multiple departments.

Includes: Joint planning · Approvals · Programme governance

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

23 Cross-Sector Policy & Regulatory Alignment

Whether the State has the capability to digitally support alignment of policies, regulations, standards and institutional responsibilities across sectors.

Includes: Policy alignment · Regulatory coordination

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

State Functionaries

Cross-department coordination and strategic decision-making

24 Strategic Intelligence & Decision Support

Whether the State has the capability to provide integrated analytics, dashboards and cross-sector insights that enable evidence-based planning, monitoring and strategic decision-making.

Includes: Analytics · Dashboards · Performance monitoring

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

25 Emergency Preparedness & Resilience Management

Whether the State has the capability to digitally coordinate preparedness, emergency response, disaster management and recovery across departments to ensure continuity of drinking water services.

Includes: Disaster management · Emergency response · Drought · Floods · Contamination events

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Layer 4 subtotal — State Functionaries

Sum of 5 ratings

Score: _____ / 20

Shared Digital Services

Services shared across all stakeholder groups

26 Stakeholder Identity & Lifecycle Management

Whether the State has the capability to digitally onboard, authenticate, authorize and manage the lifecycle, profiles and access of all stakeholders across the water service ecosystem.

Includes: User management · Onboarding · Identity · Roles · Access

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

27 Capacity Building & Knowledge Management

Whether the State has the capability to digitally deliver training, certifications, knowledge resources, SOPs and continuous learning opportunities to strengthen stakeholder competencies.

Includes: LMS · Training · Certification · Knowledge · Digital learning

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

28 Personalized Dashboards & Reports

Whether the State has the capability to provide stakeholders with role-based dashboards, reports, alerts and insights that support operational monitoring, performance management and informed decision-making.

Includes: Role-based dashboards · Personalized KPIs · Drill-down · Self-service reporting · Automated distribution

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Shared Digital Services

Services shared across all stakeholder groups

29 Intelligent Assistance & Decision Support

Whether the State has the capability to provide contextual assistance, conversational interfaces, recommendations and guided workflows that help stakeholders perform tasks more effectively.

Includes: Context-aware guidance · Decision support · Recommendations · Conversational assistance

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

30 Omnichannel Digital Experience

Whether the State has the capability to enable stakeholders to access digital services through multiple channels such as mobile applications, web portals, WhatsApp, SMS, IVRS and future digital interfaces.

Includes: WhatsApp · Mobile · Portal · IVRS · SMS · Email

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Layer 5 subtotal — Shared Digital Services

Sum of 5 ratings

Score: _____ / 20

Technology Foundation

The software layer underlying every other layer

31 Digital Platform & Architecture

Whether the State has the capability to build and operate scalable, modular, resilient and maintainable digital solutions using modern architectural principles, reusable components and cloud-ready technologies.

Includes: Unified platform · Modular architecture · Microservices · Cloud native · DevOps · Containerization

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

32 Interoperability & Open Integration

Whether the State has the capability to seamlessly exchange information and integrate with central, state and external systems through open standards, APIs, common data models and interoperable interfaces.

Includes: APIs · Centre systems · State systems · Standards · Registries · Eventing

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

33 Data Management & Governance

Whether the State has the capability to establish trusted, high-quality and well-governed data through standardized data models, master data management, metadata, data quality and data governance practices.

Includes: Master data management · Metadata · Data quality · Data governance · Data catalog · Lineage · Stewardship

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Technology Foundation

The software layer underlying every other layer

34 Digital Identity & Access Management

Whether the State has the capability to securely manage digital identities, authentication, authorization and access control for users, applications and services while ensuring privacy and accountability.

Includes: IAM · SSO · RBAC · MFA

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

35 Intelligent Automation & Decision Support

Whether the State has the capability to leverage intelligent technologies to automate business processes, generate insights, support decision-making and improve operational efficiency.

Includes: AI · ML · OCR · Prediction · Recommendations · NLP

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

36 Security, Privacy & Compliance

Whether the State has the capability to protect digital systems and data through comprehensive security controls, privacy safeguards, risk management and compliance with applicable standards and regulations.

Includes: Security · Privacy · Compliance · Audit · Encryption · Monitoring

☐ **0** Does not exist ☐ **1** Under development ☐ **2** Pilot complete ☐ **3** Functional at limited scale ☐ **4** Fully functional at state scale

Evidence / notes

Layer 6 subtotal — Technology Foundation

Sum of 6 ratings

Score: _____ / 24

Assessment summary

Carry each layer subtotal here, total them, then read the maturity index below.

LAYER	SUBTOTAL	MAXIMUM
1. Citizens		24
2. Frontline Workers		24
3. Department		32
4. State Functionaries		20
5. Shared Digital Services		20
6. Technology Foundation		24
Total		144

Overall maturity index

Total ÷ 144 × 100 = %

0-20%	Nascent
21-40%	Emerging
41-60%	Developing
61-80%	Mature
81-100%	Leading

Layer maturity index

Read each layer subtotal out of its maximum above.

0-20%	Nascent
21-40%	Emerging
41-60%	Developing
61-80%	Mature
81-100%	Leading

Notes · evidence · systems referenced